

SERIES

```
import pandas as pd
a=pd.series([1,2,3,4,6])
print(a)
```

```
-----
-----
AttributeError                                Traceback (most recent call
last)
```

```
Cell In[3], line 2
      1 import pandas as pd
----> 2 a=pd.series([1,2,3,4,6])
      3 print(a)
```

```
AttributeError: module 'pandas' has no attribute 'series'
```

```
!pip install pandas
```

```
Requirement already satisfied: pandas in c:\users\monik\appdata\
roaming\python\python313\site-packages (3.0.0)
Requirement already satisfied: numpy>=1.26.0 in c:\users\monik\
appdata\roaming\python\python313\site-packages (from pandas) (2.4.2)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\
monik\appdata\roaming\python\python313\site-packages (from pandas)
(2.9.0.post0)
Requirement already satisfied: tzdata in c:\users\monik\appdata\
roaming\python\python313\site-packages (from pandas) (2025.3)
Requirement already satisfied: six>=1.5 in c:\users\monik\appdata\
roaming\python\python313\site-packages (from python-dateutil>=2.8.2-
>pandas) (1.17.0)
```

```
import pandas as pd
a=pd.Series([1,2,3,4,6])
print(a)
print(a.values)
print(a.index)
print(a.dtype)
```

```
0    1
1    2
2    3
3    4
4    6
dtype: int64
[1 2 3 4 6]
RangeIndex(start=0, stop=5, step=1)
int64
```

```
import pandas as pd
a=pd.Series([True,False,True])
print(a)
print(a.values)
print(a.index)
print(a.dtype)

0    True
1   False
2     True
dtype: bool
[ True False  True]
RangeIndex(start=0, stop=3, step=1)
bool
```

```
import pandas as pd
a=pd.Series([True,3,4,6])
print(a)
print(a.values)
print(a.index)
print(a.dtype)

0    True
1      3
2      4
3      6
dtype: object
[True 3 4 6]
RangeIndex(start=0, stop=4, step=1)
object
```

```
import pandas as pd
a=pd.Series(['Monika','devashree','varsshini','sanjeevi'])
print(a)
print(a.values)
print(a.index)
print(a.dtype)
#accessing element
print(a[1])
#accessing element using loc
print(a.loc[1])
#accessding element using iloc
print(a.iloc[0])
#slicing
print(a[1:3])
print(a.loc[1:3])
print(a.iloc[1:3])

0    Monika
1   devashree
```

```

2    varsshini
3    sanjeevi
dtype: str
<ArrowStringArray>
['Monika', 'devashree', 'varsshini', 'sanjeevi']
Length: 4, dtype: str
RangeIndex(start=0, stop=4, step=1)
str
devashree
devashree
Monika
1    devashree
2    varsshini
dtype: str
1    devashree
2    varsshini
3    sanjeevi
dtype: str
1    devashree
2    varsshini
dtype: str

import pandas as pd
arr = pd.Series([12,23,34,45,56],index=['A','B','C','D','E'])
a=pd.Series(['monika','Dharunya','Sreya'],index=['Name1','Name2','Name3'])
print(a)
print(a['Name1'])
print(a+ 'sri')
print(arr)
print(arr['D'])
#access all the elemnet instance of creating loop'
print(arr+5)
print(arr.iloc[1])
print(arr.loc['A'])
#Statistical function
print(arr.sum())
print(arr.std())
print(arr.max())
print(arr.min())
print(arr.mean())
print(arr.median())
#boolean indexing
print(arr[arr>30])
print(len(arr))
print(len(a))
#individuall value count
print(arr.value_counts())
#sring operarion
name=pd.Series(['Jack','vishnu','anitha','sasmitha','srinivas'])

```

```

print(name.str.upper())
print(name.str.len())
print(len(name))
print(arr[['A','C','E']])

```

```

Name1      monika
Name2      Dharunya
Name3      Sreya

```

```
dtype: str
```

```
monika
```

```

Name1      monikasri
Name2      Dharunyasri
Name3      Sreyasri

```

```
dtype: str
```

```
A      12
```

```
B      23
```

```
C      34
```

```
D      45
```

```
E      56
```

```
dtype: int64
```

```
45
```

```
A      17
```

```
B      28
```

```
C      39
```

```
D      50
```

```
E      61
```

```
dtype: int64
```

```
23
```

```
12
```

```
170
```

```
17.392527130926087
```

```
56
```

```
12
```

```
34.0
```

```
34.0
```

```
C      34
```

```
D      45
```

```
E      56
```

```
dtype: int64
```

```
5
```

```
3
```

```
12      1
```

```
23      1
```

```
34      1
```

```
45      1
```

```
56      1
```

```
Name: count, dtype: int64
```

```
0      JACK
```

```
1      VISHNU
```

```
2      ANITHA
```

```

3     SASMITHA
4     SRINIVAS
dtype: str
0     4
1     6
2     6
3     8
4     8
dtype: int64
5
A     12
C     34
E     56
dtype: int64

import pandas as pd
a=pd.Series({
    "Gayathri":16,
    "Rithanya":37,
    "bavanthika":8})
print(a)
print(a["Rithanya"])

Gayathri      16
Rithanya      37
bavanthika     8
dtype: int64
37

s1=pd.Series([10,20,30],index=['A','B','C'])
s2=pd.Series([1,2,3],index=['B','C','D'])
print(s1+s2)
s=pd.Series([1,2,3,4,4,5,5])
print(s.count())
print(s.unique())
print(s.nunique())

A     NaN
B     21.0
C     32.0
D     NaN
dtype: float64
7
[1 2 3 4 5]
5

```

DATA FRAME

```
import pandas as pd
#method1 : from dictionaty
data = {
    'name':['Monika','nagul','prem'],
    'rollno':[27,20,24],
    'Age':[19,20,21]}
df=pd.DataFrame(data)
print(df)
#from series to dataframe
s1=pd.Series([1,2,3,4,5])
s2=pd.Series([5.6,6.7,8.9,88.9])
s3=pd.Series([True,False,True,False])
df=pd.DataFrame({'s.no':s1,
                 'cgpa':s2,
                 'attendance':s3
})
print(df)
#list of list
data=[['athish',22,98],
      ['Rithika',20,89],
      ['Anu',87,90]]
df=pd.DataFrame(data,columns=["Name","Age","Mark1"])
print(df)
#List of distionary value
data=[
    {"name":"Sakthi","course":"Data Science","grade":'A'},
    {"name":"ram","course":"AIMML","grade":'A+'},
    {"name":"San","course":"AIML","grade":'A++'}]
df=pd.DataFrame(data,index=["std1","std2","std3"])
print(df)
#adding one extra column
df['mark']=[98,87,67]
print(df)
#adding one extra row
# adding one extra row
df.loc['std4'] = ["Mani", "Data Analytics", "A", 95]
print(df)
std5=pd.DataFrame([{"name":"Sau","course":"AI","grade":'A++'}])
df=pd.concat([df,std5])
print(df)
#create multiple row
students=pd.DataFrame([{"name":"sanjeevi","course":"csda","grade":"0+",
    "mark":90},
    {"name":"divya","course":"aiml","grade":"0+","mark":99}],index=["std5",
    "std6"])
df=pd.concat([df,students])
```

```

print(df)
#column aaccess by bracket notation and dot notation
print(df.course)
print(df[['grade','name']])
print(df.shape)
#find number of row
print(df.shape[0])
#find number of columns
print(df.shape[1])
#show all the data types of all rows ans columns
print(df.dtypes)
#info
print(df.info)
#describe
print(df.describe)
#select specific row
print(df.loc[['std1','std3']])
#update value
df.loc['std3','grade']='B++'
print(df)
#delete column and del:
df.drop('grade',axis=1,inplace=True)
print(df)
#select row and del
df.drop('std5',axis=0,inplace=True)
print(df)
#sorting
#df.sort_values('marks',inplace=True)
#print(df)
#df.sort_values('mark',ascending=False,inplace=True)
#print(df)
#statistical function for particular column
#print(df["marks"].max())
#print(df)

```

	name	rollno	Age
0	Monika	27	19
1	nagul	20	20
2	prem	24	21

	s.no	cgpa	attendance
0	1	5.6	True
1	2	6.7	False
2	3	8.9	True
3	4	88.9	False
4	5	NaN	NaN

	Name	Age	Mark1
0	athish	22	98
1	Rithika	20	89
2	Anu	87	90

	name	course	grade
--	------	--------	-------

```

std1  Sakthi  Data Science      A
std2   ram      AIMML      A+
std3   San      AIML       A++
      name      course grade  mark
std1  Sakthi  Data Science      A    98
std2   ram      AIMML      A+    87
std3   San      AIML       A++    67
      name      course grade  mark
std1  Sakthi    Data Science      A    98
std2   ram      AIMML      A+    87
std3   San      AIML       A++    67
std4   Mani  Data Analytics      A    95
      name      course grade  mark
std1  Sakthi    Data Science      A  98.0
std2   ram      AIMML      A+  87.0
std3   San      AIML       A++  67.0
std4   Mani  Data Analytics      A  95.0
0      Sau      AI       A++   NaN
      name      course grade  mark
std1  Sakthi    Data Science      A  98.0
std2   ram      AIMML      A+  87.0
std3   San      AIML       A++  67.0
std4   Mani  Data Analytics      A  95.0
0      Sau      AI       A++   NaN
std5  sanjeevi      csda      0+  90.0
std6   divya      aiml      0+  99.0
std1      Data Science
std2      AIMML
std3      AIML
std4      Data Analytics
0      AI
std5      csda
std6      aiml
Name: course, dtype: str
      grade      name
std1      A      Sakthi
std2     A+       ram
std3     A++      San
std4      A      Mani
0      A++      Sau
std5     0+  sanjeevi
std6     0+   divya
(7, 4)
7
4
name      str
course    str
grade     str
mark      float64

```



```

~\AppData\Local\Temp\ipykernel_8300\1472674708.py in ?()
    68 #select row and del
    69 df.drop('std5',axis=0,inplace=True)
    70 print(df)
    71 #sorting
--> 72 df.sort_values('marks',inplace=True)
    73 print(df)
    74 #df.sort_values('mark',ascending=False,inplace=True)
    75 #print(df)

```

```

~\AppData\Roaming\Python\Python313\site-packages\pandas\core\frame.py
in ?(self, by, axis, ascending, inplace, kind, na_position,
ignore_index, key)
    8343         )
    8344         elif by:
    8345             # len(by) == 1
    8346
-> 8347             k = self._get_label_or_level_values(by[0],
axis=axis)
    8348
    8349             # need to rewrap column in Series to apply key
function
    8350             if key is not None:

```

```

~\AppData\Roaming\Python\Python313\site-packages\pandas\core\
generic.py in ?(self, key, axis)
    1772         values = self.xs(key,
axis=first_other_axes)._values
    1773         elif self._is_level_reference(key, axis=axis):
    1774             values =
self.axes[axis].get_level_values(key)._values
    1775         else:
-> 1776             raise KeyError(key)
    1777
    1778         # Check for duplicates
    1779         if values.ndim > 1:

```

KeyError: 'marks'

```

import pandas as pd
#read csv file
df=pd.read_csv('people-100.csv',usecols=["Last Name","Sex"])
print(df)
#read only 1st 5 data
print(df.head())
#read only last file
print(df.tail())
print(df.to_string())
#select specific column

```

```
print(df['Sex'])
```

	Last Name	Sex
0	Terrell	Male
1	Summers	Female
2	Travis	Male
3	Martinez	Male
4	Todd	Male
..	...	...
95	Barnes	Female
96	Patterson	Female
97	Bray	Male
98	Oconnell	Female
99	Bernard	Male

```
[100 rows x 2 columns]
```

	Last Name	Sex
0	Terrell	Male
1	Summers	Female
2	Travis	Male
3	Martinez	Male
4	Todd	Male
	Last Name	Sex
95	Barnes	Female
96	Patterson	Female
97	Bray	Male
98	Oconnell	Female
99	Bernard	Male
	Last Name	Sex
0	Terrell	Male
1	Summers	Female
2	Travis	Male
3	Martinez	Male
4	Todd	Male
5	Day	Male
6	Buck	Female
7	Hinton	Male
8	Farrell	Male
9	Downs	Male
10	Ross	Female
11	Newton	Male
12	Blake	Male
13	Nash	Female
14	Farmer	Male
15	Knight	Female
16	Rangel	Male
17	Meadows	Male
18	Paul	Male
19	Mitchell	Female

20	Norton	Female
21	Hatfield	Female
22	Barnes	Male
23	Rhodes	Female
24	Mcneil	Female
25	Levine	Male
26	Payne	Male
27	Vega	Female
28	Buckley	Male
29	Weber	Male
30	Camacho	Female
31	Gallagher	Male
32	Andrews	Female
33	Schmidt	Female
34	Sparks	Female
35	Cook	Male
36	Hurst	Male
37	Rich	Female
38	Wiggins	Female
39	Mckinney	Female
40	Stone	Female
41	Walsh	Female
42	Vang	Female
43	Hill	Female
44	Andrews	Male
45	Pugh	Female
46	Estrada	Female
47	Brady	Male
48	Frey	Male
49	Sweeney	Male
50	Escobar	Female
51	Oconnell	Female
52	Esparza	Female
53	Rivas	Female
54	Payne	Female
55	Rocha	Female
56	Watkins	Female
57	Olsen	Female
58	Nelson	Female
59	Phelps	Male
60	Daugherty	Male
61	Garner	Male
62	Bennett	Male
63	Conway	Female
64	Barnett	Female
65	Harper	Male
66	Ward	Female
67	Rubio	Female
68	Pineda	Male

69	Wu	Male
70	Benton	Male
71	Hull	Male
72	Ritter	Female
73	Briggs	Female
74	Andrews	Female
75	Cunningham	Male
76	Atkinson	Male
77	Reid	Female
78	Stevens	Male
79	Huff	Male
80	Foley	Female
81	Waters	Male
82	Stephenson	Male
83	Reese	Female
84	Williamson	Female
85	Vaughn	Male
86	Dickson	Male
87	Bailey	Male
88	Medina	Female
89	Haney	Female
90	Nixon	Female
91	Hardy	Female
92	Cox	Female
93	Cole	Male
94	Guzman	Female
95	Barnes	Female
96	Patterson	Female
97	Bray	Male
98	Oconnell	Female
99	Bernard	Male
0	Male	
1	Female	
2	Male	
3	Male	
4	Male	
	...	
95	Female	
96	Female	
97	Male	
98	Female	
99	Male	

Name: Sex, Length: 100, dtype: str